

Distal Edge Verification in Proton Therapy Using An In-Room CRYSP Scanner

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Abstract

The range of a therapeutic proton beam is the quantity a treatment plan controls least well, and the distal edge of the field is where that uncertainty is paid for clinically [1]. The β^+ activity induced in tissue by the beam carries an image of where the protons stopped, and an in-room PET scanner can record it [2]. This note describes CryspBrainSim (CBS), a simulation and analysis chain that starts from the list-mode lines of response a PET scanner would register for a spread-out Bragg peak (SOBP) field, reconstructs the β^+ activity by maximum-likelihood expectation maximisation (MLEM) [3], and extracts the distal endpoint R of the activity distribution together with its statistical precision σ_R . The reference case is an ependymoma at the brainstem interface, modelled as a 4×4 cm target at 5.7 cm depth inside a uniform brain-tissue ellipsoid, irradiated to 1 Gy. The reference scanner is a CRYSP cryogenic PET [4] with an inner radius of 387 mm, an axial field of view of 1024 mm and 960 monolithic BGO crystals. The depth-activity profile is integrated over the whole transverse plane and its distal falloff fitted with a complementary error function, from which the half-height point R_{50} (the precision observable) and the practical endpoint R_p (the accuracy observable) follow analytically. On a single full-statistics acquisition the reconstructed endpoint reproduces the endpoint of the generated activity to better than one voxel, and the ensemble spread of the ten independent acquisitions gives $\sigma_R = 0.065$ mm at 1 Gy — a statistical precision far below the millimetre scale at which range uncertainty is currently budgeted. The same chain, with the grid, the analysis window, the fit model and the estimator held fixed, is the instrument used to compare competing scanner geometries.

1 Introduction

CryspBrainSim (CBS) fits the activity distribution of the β^+ emitters produced in the decay of the various isotopes activated by a proton beam crossing (brain) tissue and delivering a conformal dose into a (head) tumour. The case chosen as representative is an ependymoma located near the brainstem (figure 1).

The fit to the β^+ activity allows the determination of the distal edge of the proton beam (R), which is the critical measurement to be pinned down in proton beam treatment [5]. The goal is to minimise the *bias* of the R determination (that is, the difference between R and the true distal edge D), as well as σ_R (the r.m.s. spread of R over an ensemble of acquisitions).

The input to CBS is the list of lines of response (LORs) produced by a PET scanner specified by a number of parameters, including PET diameter, axial field of view (AFOV), crystal dimensions and crystal type. For each scanner a sensitivity image is computed, and MLEM [3] is used to reconstruct the activity image A . R is estimated through a depth-profile extraction and an endpoint estimation, as described below.

2 The phantom

The phantom (figure 2) describing the study case is an ellipsoid approximating the shape of a head. The beam enters along the z axis, and the target region, marked in the figure, is a 4×4 cm box located at a depth of 5.7 cm from the beam entry point. For simplicity a uniform phantom, filled with brain tissue (G4_BRAIN_ICRP), is used.

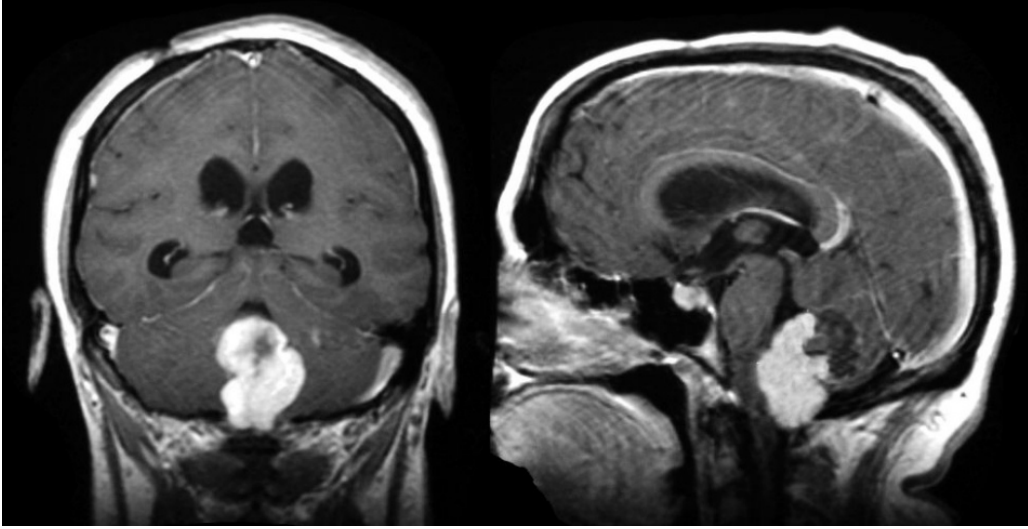


Figure 1: Gadolinium-enhanced T1-weighted MRI of a fourth-ventricle ependymoma. (*Left*) coronal and (*right*) sagittal views show a large midline enhancing mass invading the roof of the fourth ventricle and growing downward toward the foramen magnum, with the brainstem displaced anteriorly. The tumour–brainstem interface visible here is precisely the region where the distal edge of a proton beam must be placed. Image credit: <https://case.edu/med/neurology/NR/cerebelepend2.htm>.

3 The beam

The proton beam delivers a typical dose of 1 Gy. Figure 3 shows the emitter trail by isotope along with the isotope mix and the production profiles. Figure 4 shows the activity of those isotopes. Figure 5 shows the spread-out Bragg peak (SOBP), which delivers a conformal dose to the target region (left panel), as well as the relation between dose and activity (right panel).

4 The scanner

The reference scanner is a CRYSP¹ model, with an inner radius $R_{\text{inner}} = 387$ mm and an AFOV of 1024 mm. It deploys monolithic BGO crystals of 50×50 mm² surface and a depth of 3.3 radiation lengths ($3.3 X_0$), or 37 mm. The scanner has 20 wheels and 48 crystals per wheel, for a total of 960 crystals. The energy resolution of the scanner is 10% at 511 keV and the spatial resolution in the three coordinates is 3.5 mm FWHM. The coincidence time window for the LORs is 3 ns [4].

Figure 6 shows the detected (blue histogram) versus true (green histogram) activity in the scanner. The detected activity is strongly biased by absorption in the phantom. Figure 7 shows the reconstructed (blue histogram) versus true (green histogram) activity profile in the scanner, together with the fit to the profile.

The fit to a single run yields a predicted distal value of -16.10 mm, to be compared with the distal value of the true beam profile, -16.45 mm.

5 Depth-profile extraction and endpoint estimation

Let $\lambda(\mathbf{r})$ denote the reconstructed activity image, resampled onto a beam-aligned grid with $\mathbf{r} = (x, y, z)$ and the beam (depth) along z . The one-dimensional depth–activity profile is the

¹CRYogenic Sensitive PET

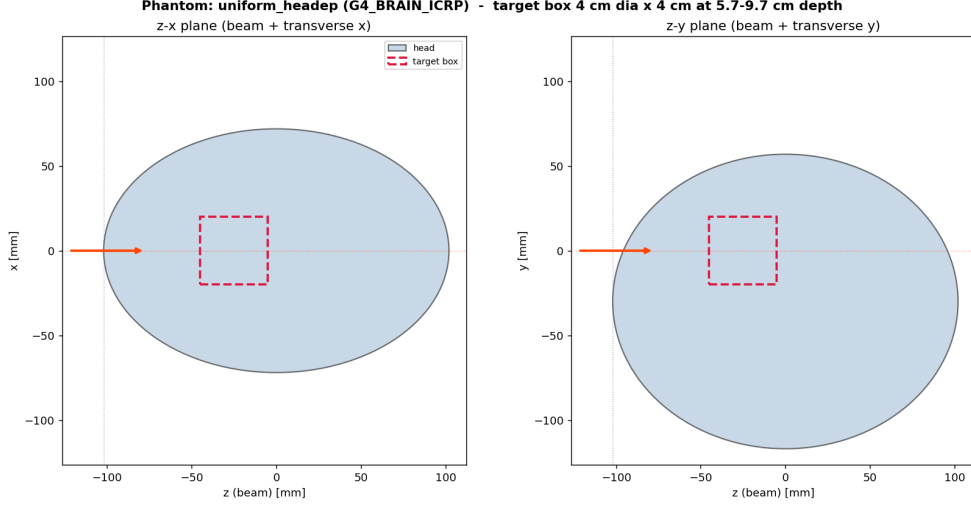


Figure 2: The phantom describing the study case is an ellipsoid filled with uniform brain tissue.

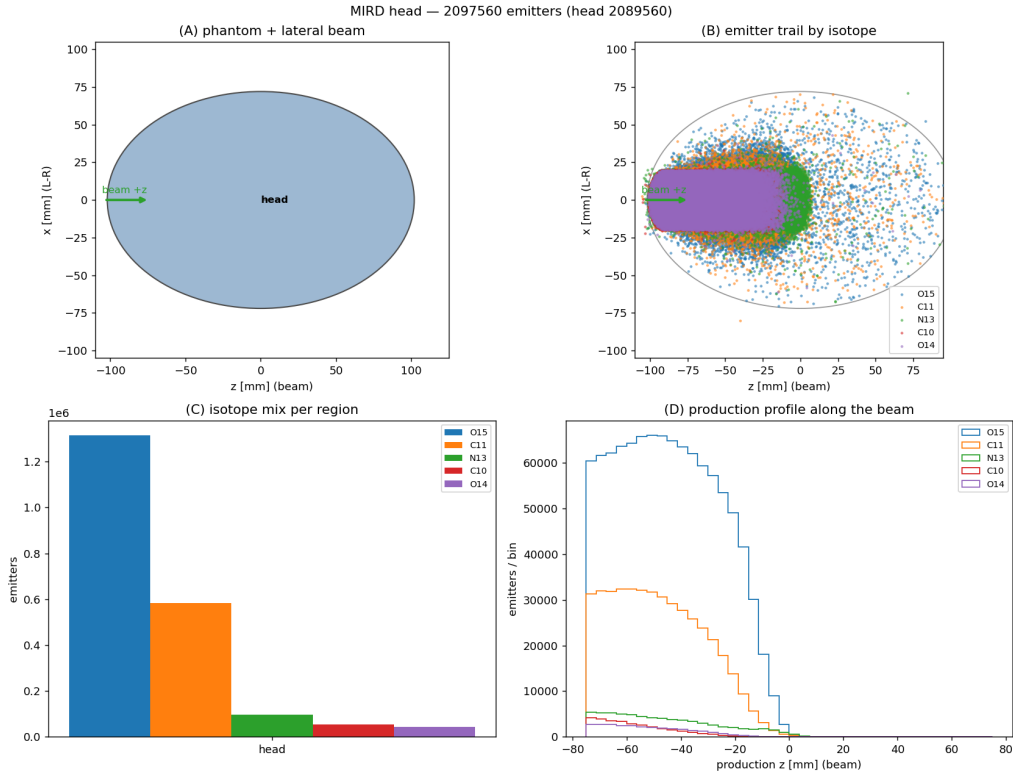


Figure 3: a) A proton beam enters the phantom moving along the z axis; b) β^+ emitter trail by isotope; c) isotope mix; d) isotope production profile along the beam.

transverse integral over the full image plane,

$$P(z) = \int \lambda(x, y, z) dx dy \approx \Delta x \Delta y \sum_{i,j} \lambda_{ijk}(z), \quad (1)$$

carried over the whole transverse extent of the reconstruction grid. Integrating the full plane makes $P(z)$ the same observable for the reconstructed image and for the generated activity, and keeps the endpoint independent of the transverse beam envelope: the halo produced by multiple Coulomb scattering widens with depth, so a fixed transverse aperture captures a depth-decreasing

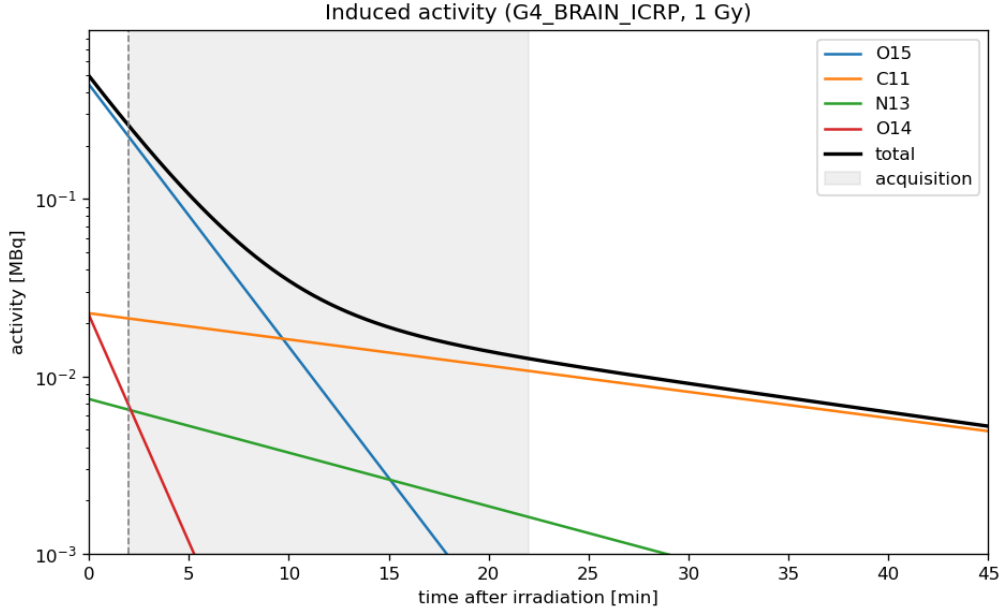


Figure 4: Activity of the isotopes producing the β^+ trail.

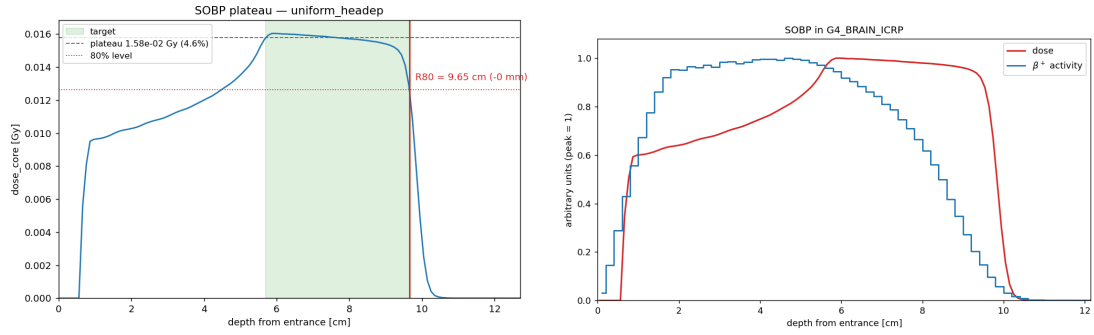


Figure 5: Left: the SOBP beam and the definition of R_{80} . Right: the (nominal) activity distribution, together with the beam profile.

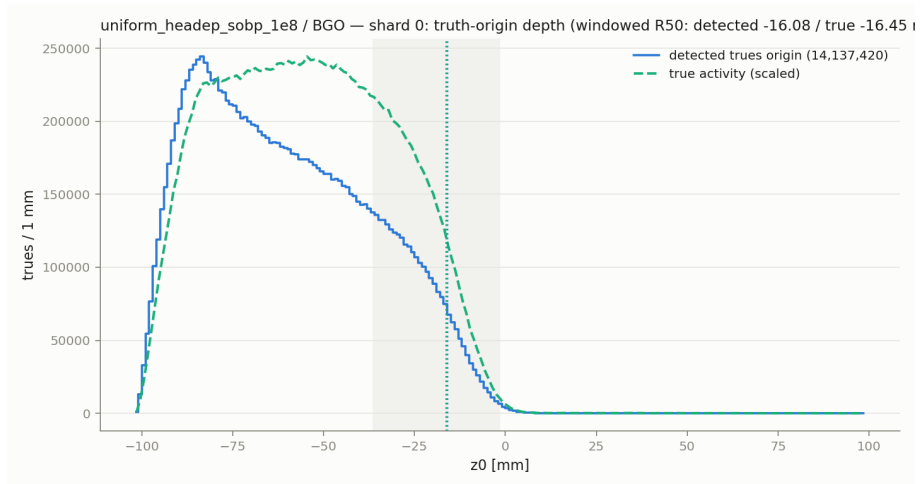


Figure 6: Detected (blue histogram) versus true (green histogram) activity in the scanner. The detected activity is strongly biased due to absorption.

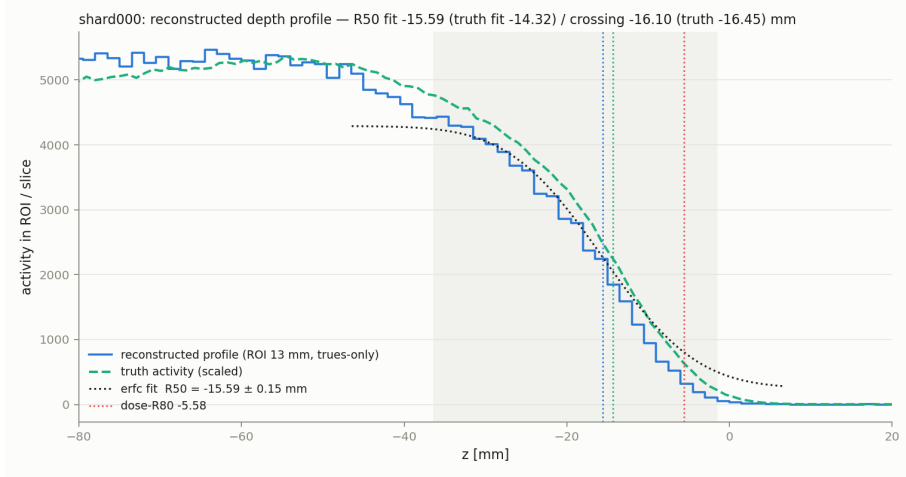


Figure 7: Corrected (blue histogram) versus true (green histogram) activity in the scanner, together with the fit to the profile.

fraction of the activity and displaces the fitted endpoint proximally (by 2 mm to 3 mm for a disc of 13 mm radius in the geometry of this study). Equation (1) is an integration (a sum), never an average or a normalisation: summing preserves the counting statistics on which the endpoint fit and σ_R depend. Because $P(z)$ is a sum of voxel counts it is, to first approximation, Poisson with variance $\text{Var}[P(z)] \approx \bar{P}(z)$; the MLEM reconstruction correlates neighbouring voxels, so this is a working approximation used only to weight the fit, while the authoritative endpoint uncertainty is the ensemble spread defined below.

The activity profile is not a single sigmoid—it rises from the entrance, plateaus, and falls at the distal edge, which lies *upstream* of the Bragg peak because the β^+ -production channels close below their reaction thresholds. The endpoint is therefore estimated within a *fixed* distal analysis window $\mathcal{W} = [z_0^{\text{nom}} - \delta_-, z_0^{\text{nom}} + \delta_+]$, bracketing the falloff and defined from the *nominal* range, identically for every arm and realisation; a per-realisation peak search would correlate $\mathcal{W}$ with the very fluctuations being measured. Inside $\mathcal{W}$ the falloff is modelled by a complementary error function,

$$\hat{P}(z) = b + a \frac{1}{2} \text{erfc}\left(\frac{z - z_0}{\sqrt{2} w}\right), \quad (2)$$

so that z_0 is by construction the R_{50} point — the same activity-edge observable, fitted on a whole-plane profile, that Zapien-Campos *et al.* [6] adopt as the PET-activity range parameter — a and b absorb the plateau height and background (making z_0 invariant to the absolute activity scale), and w is the edge width. Any falloff level follows analytically,

$$R_x = z_0 + \sqrt{2} w \text{erfc}^{-1}(2x), \quad (3)$$

with R_{50} the most precisely determined point (the steepest slope of the edge) and R_{80}, R_{20} as tail diagnostics. The parameters are obtained by minimising the Poisson-weighted residual

$$\chi^2 = \sum_{z \in \mathcal{W}} \frac{[P(z) - \hat{P}(z)]^2}{\max(P(z), 1)}, \quad (4)$$

which yields the parameters $z_0^{(k)}, w^{(k)}$ and their covariance for each realisation k .

The predictor of the beam endpoint is the *practical endpoint*, obtained by extrapolating the tangent of the fitted edge at z_0 down to the baseline,

$$R_p = z_0 + \sqrt{\frac{\pi}{2}} w, \quad (5)$$

the construction used for the practical range of electron beams; on the fitted edge it marks the point where the signal has fallen to 10.5% of the plateau, and its uncertainty follows from the (z_0, w) covariance. The extrapolated crossing of the activity towards zero is reported at a fixed small fraction of the plateau through Eq. (3) ($R_{1\%} = z_0 + 2.326 w$); it lies distal of R_p , and since it sits far from the steep part of the edge its uncertainty is dominated by that of the width w . R_p therefore serves as the primary predictor and $R_{1\%}$ as a tail diagnostic. Both constructions, applied identically to the depth-dose curve, give the endpoint-to-endpoint distance between the β^+ activity and the delivered dose.

Finally, over the ensemble of K realisations at a given delivered dose, the mean endpoint estimates the systematic activity-edge position while its spread is the terminal figure of merit,

$$\bar{R}_p = \frac{1}{K} \sum_k R_p^{(k)}, \quad \sigma_R = \sqrt{\frac{1}{K-1} \sum_k (R_p^{(k)} - \bar{R}_p)^2} \sim \frac{1}{\sqrt{N}}, \quad (6)$$

recovering the counts-limited scaling in the detected-coincidence number N . The methodological invariant of the comparison is that the grid, the window $\mathcal{W}$, the model (2) and the estimator (4) are identical across the three geometries: the *only* quantity that differs entering σ_R is the reconstructed image content. This is also where the open dual-head geometry is tested most sharply—its limited-angle artefact elongates the activity distribution, and whenever that elongation projects onto the beam (z) axis it biases R_p directly, so σ_R is reported as a function of beam-axis orientation relative to the head gap.

References

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